Chapter 6 — Dictionaries, Sets and the `collections` Toolbox

Lists and tuples find things by **position**. Dictionaries and sets find things by **value** — and they do it in constant time on average, whether they hold ten items or ten million. That single property makes them the workhorses of real Python code: caches (stores of already-computed results, reused to save time — see System Design Fundamentals), indexes (lookup tables pointing to where data lives), counters, configuration (a program's settings), JSON (the text data format web APIs use), de-duplication (removing repeats), graph algorithms (algorithms over networks of connected points, such as route finding). This chapter explains how they work (hashing — turning a value into an integer that says where to store it), how to use every method, and which specialised container from `collections`, `heapq` and `bisect` to reach for when a plain dict, set or list isn't quite right.

> **Key term — Big-O notation:** this chapter measures costs with Big-O, which describes how the work grows as the number of items *n* grows. **O(1)** ("constant time") means the cost does not depend on size; **O(n)** ("linear") means it grows in proportion to size; **O(log n)** grows very slowly (doubling *n* adds just one step). It describes the trend, not exact timings.

What you'll learn

1. Dictionaries: define, access, update, remove, copy, loop, operators (`in`, `|`), and insertion order.
2. Hashing: what "hashable" means, how a dict/set finds a key, and the big-O table.
3. Nested dictionaries and their link to JSON.
4. Every dict method, with `get`, `setdefault`, `update` and `|` compared, plus view objects.
5. Dict and set comprehensions.
6. Sets: define, add/update, remove/discard, copy, loop.
7. Set operations (union, intersection, difference, symmetric difference) and every set method.
8. `frozenset`.
9. `collections`: `Counter`, `defaultdict`, `OrderedDict`, `deque`, `ChainMap`, `namedtuple`.
10. `heapq` (priority queues — the most urgent item always comes out first) and `bisect` (binary search — finding a position by repeatedly halving a sorted list), and a when-to-use table for everything.

All runnable code is in `code/`. Outputs are real (Python 3.14; code runs on 3.12+). Where a set of **strings** is printed, its order can differ between runs — string hashes are randomised per process (a *process* is one running instance of a program; explained in 6.2).

6.1 Dictionaries — key → value

What it is

A **dictionary** (`dict`) stores **key–value pairs**. You look up a value by its key, not by position: `ages["Alice"]`. Keys are unique; values can be anything (numbers, strings, lists, other dicts).

- Written with braces and colons: `{"Alice": 22, "Bob": 25}`.
- **Mutable:** add, change and delete pairs at any time.
- **Keys must be hashable** (roughly: immutable — strings, numbers, tuples of hashables, `None`, `frozenset`). Lists, dicts and sets can't be keys. See 6.2.
- **Insertion-ordered:** iteration returns keys in the order they were first inserted. This is a language guarantee since **Python 3.7** (it was an implementation detail — true of CPython, but not

promised by the language — in CPython 3.6). Older tutorials that call dicts "unordered" are out of date — but dicts are still not *indexable* by position (`d[0]` looks up the key `0`).

Analogy: a real dictionary — you jump straight to a word and read its definition; you don't scan page by page.

Why it matters

Whenever you'd otherwise search a list for "the item whose name is X", a dict does it in one step. Python itself runs on dicts: module globals (a file's top-level variables), object attributes (`obj.__dict__`), keyword arguments and class namespaces (tables mapping names to objects) are all dicts.

How it works — define and access

```
ages = {"Alice": 22, "Bob": 25, "Eve": 19}
print(ages, len(ages))
print(ages["Alice"], ages.get("Bob"), ages.get("Charlie"), ages.get("Charlie", "Not Found"))
try:
    ages["Charlie"]
except KeyError as e:
    print("KeyError:", e)
other_ways = dict(a=1, b=2), dict([("x", 1), ("y", 2)]), dict(zip("pq", [7, 8]))
print(*other_ways)
print({(59.9, 10.7): "Oslo", 42: "int key", None: "None key"})
try:
    {["a", "list"]: 1}
except TypeError as e:
    print("TypeError:", e)
```

Output:

```
{'Alice': 22, 'Bob': 25, 'Eve': 19} 3
22 25 None Not Found
KeyError: 'Charlie'
{'a': 1, 'b': 2} {'x': 1, 'y': 2} {'p': 7, 'q': 8}
{(59.9, 10.7): 'Oslo', 42: 'int key', None: 'None key'}
TypeError: cannot use 'list' as a dict key (unhashable type: 'list')
```

- `d[key]` raises `KeyError` if the key is missing — use it when a missing key is a bug.
- `d.get(key)` returns `None` (or your default) instead — use it when absence is normal.
- Four ways to build: literal, `dict(k=v)` (string keys only), `dict(list_of_pairs)`, `dict(zip(keys, values))`.

How it works — update, remove, copy, loop

```
ages = {"Alice": 22, "Bob": 25, "Eve": 19}
ages["Alice"] = 23                          # overwrite
ages["Dave"] = 20                           # insert (goes to the end)
print(ages)
removed = ages.pop("Eve");  print("pop:", removed, ages)
del ages["Bob"];            print("del:", ages)
print(ages.pop("Zed", "no such key"))       # default avoids KeyError
```

Output:

```
{'Alice': 23, 'Bob': 25, 'Eve': 19, 'Dave': 20}
pop: 19 {'Alice': 23, 'Bob': 25, 'Dave': 20}
del: {'Alice': 23, 'Dave': 20}
no such key
```

Assignment `d[k] = v` either **updates** (key exists — its position in the order is kept) or **inserts** (new key — goes to the end). `pop` returns the removed value; `del` just deletes.

```python
original = {"Alice": 23, "Charlie": 21, "tags": ["x"]}
copied = original.copy()                    # shallow
copied["Alice"] = 24
copied["tags"].append("y")                  # shared inner list!
print(original, copied, sep="\n")
for key in original:                        # iterates keys
    print(key, end=" ")
for key, value in {"Alice": 23, "Charlie": 21}.items():
    print(f"{key} is {value} years old")
```

Output:

```
{'Alice': 23, 'Charlie': 21, 'tags': ['x', 'y']}
{'Alice': 24, 'Charlie': 21, 'tags': ['x', 'y']}
Alice Charlie tags
Alice is 23 years old
Charlie is 21 years old
```

`copy()` (also `dict(d)`) is **shallow**: changing `copied["Alice"]` didn't affect the original, but both share the same `tags` list. Use `copy.deepcopy` for nested data (Chapter 5, 5.8).

Looping over a dict yields **keys**; `.values()` yields values; `.items()` yields `(key, value)` pairs, which you unpack in the `for`.

Dictionary operators

Operator	Meaning	Example
`k in d` / `k not in d`	Is `k` a **key**? (O(1))	`"Alice" in ages`
`d[k]`, `d[k] = v`, `del d[k]`	Get / set / delete	
`d1 | d2`	New merged dict; right side wins on conflicts (3.9+)	`{"a":1} | {"a":2}` → `{'a': 2}`
`d1 |= d2`	Merge into `d1` in place (like `update`)	
`{**d1, **d2}`	Merge via unpacking (3.5+)	
`==`	Same pairs (order ignored)	

```
d = {"Alice": 23, "Charlie": 21}
print("Alice" in d, "Bob" not in d, 23 in d, 23 in d.values())
a, b = {"x": 1, "y": 2}, {"y": 20, "z": 3}
print("a | b :", a | b)
a |= {"w": 0};  print("a |= :", a)
print("{**a, **b}:", {**a, **b})
print({"a": 1, "b": 2} == {"b": 2, "a": 1})
views = d.keys()
d["New"] = 0
print(views)                                # views are live
```

Output:

```
True True False True
a | b : {'x': 1, 'y': 20, 'z': 3}
a |= : {'x': 1, 'y': 2, 'w': 0}
{**a, **b}: {'x': 1, 'y': 20, 'w': 0, 'z': 3}
True
dict_keys(['Alice', 'Charlie', 'New'])
```

`23 in d` is `False` because `in` checks **keys**. To search values, use `in d.values()` — but that's an O(n) scan.

Common mistakes

- **Changing a dict's size while looping over it** → `RuntimeError: dictionary changed size during iteration`. Loop over a snapshot (a frozen copy taken now): `for k in list(d): ...`, or build a new dict with a comprehension.
- Using `d[k]` for optional keys and crashing on `KeyError`.
- Expecting `d.copy()` to copy nested lists/dicts.
- Relying on `d.keys()[0]` — views (the live windows `keys()`/`values()`/`items()` return, 6.4) aren't indexable; use `next(iter(d))`.

Real-world examples

- JSON objects map straight to dicts (`json.loads`), so every web API response you parse is a dict.
- Caches: `functools.lru_cache` (a decorator that remembers a function's results) and hand-written memo tables (tables of already-computed answers) are dicts from arguments to results.
- Django/Flask pass template context (the variables an HTML template can use) and request data as dicts (or dict-like objects).

In the interview

30-second answer: "A dict maps hashable keys to values with average O(1) get, set, delete and membership, using a hash table. Since 3.7 it preserves insertion order. `d[k]` raises KeyError, `get` returns a default, `|` merges (3.9+), and `copy()` is shallow."

- *Q: Are dicts ordered?* A: Yes, insertion order, guaranteed since 3.7. Equality still ignores order.
- *Q: What happens on `d[k] = v` for an existing key?* A: Value replaced, position unchanged.
- *Q: `in` on dict — keys or values?* A: Keys.

Self-check

1. `{"a": 1}.get("b", 0)`? → `0`.
2. `{"a": 1} | {"a": 2, "b": 3}`? → `{'a': 2, 'b': 3}`.

3. Can a tuple be a key? → Yes, if all its items are hashable.
4. How do you delete keys while looping? → Iterate over `list(d)`.

TL;DR

Key → value, O(1) average, insertion-ordered (3.7+), hashable keys, `[]` raises, `get` defaults, `|` merges, `copy` is shallow.

6.2 Hashing — how dicts and sets find things in O(1)

What it is

A **hash** is an integer computed from a value by `hash(x)`. A **hash table** is an array of slots; the hash decides which slot a key goes into. Dicts and sets are hash tables. An object is **hashable** if it has a hash that never changes during its lifetime and can be compared with `==`.

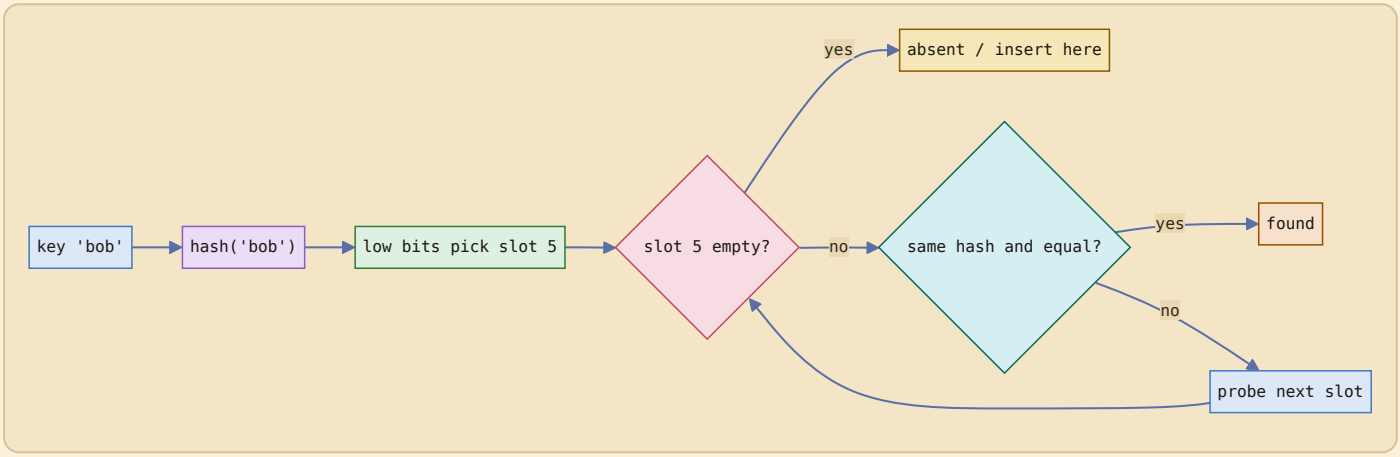
The contract (from the Python data model — the rules defining how objects plug into the language): **if `a == b` then `hash(a) == hash(b)`**. The reverse need not hold — different values may share a hash (a **collision**).

Why it matters

Hashing explains everything about dicts and sets: why lookups are O(1), why keys must be immutable, why `1` and `1.0` are the same key, and why sets have no order.

How it works — a lookup, step by step

1. Compute `h = hash(key)`.
2. Take some low bits (the last few binary digits) of `h` to choose a slot in the table (table size is a power of two).
3. If the slot is empty → key absent (for lookup) or put it here (for insert).
4. If the slot holds an entry, compare: same hash **and** (`is` or `==`) → found.
5. Otherwise it's a collision: CPython uses **open addressing** — it computes the next slot to try with a pseudo-random probe sequence (a scrambled but repeatable order of slots) (mixing in more bits of the hash), and repeats from step 3.
6. The table is kept at most about **2/3 full**. When it gets fuller, CPython allocates a larger table and re-inserts everything (**resize**). Like list growth, this is rare enough that insertion is **amortised O(1)**.



Since Python 3.6, CPython's dict is a **compact dict**: a small sparse (mostly empty) *index* table (the slots) pointing into a dense (tightly packed) *entries* array that stores `(hash, key, value)` in

insertion order. That's where insertion order comes from, and it also made dicts 20–25% smaller. Sets use a simpler, non-compact table, so they have no ordering.

Growth you can observe with `sys.getsizeof` (a table of 8 slots holds 5 entries, 16 holds 10, 32 holds 21, 64 holds 42 — the 2/3 rule):

```
 len  bytes
   1  224
   6  352
  11  632
  22  1168
  43  2264
```

What is hashable?

```
42              hashable    hash=42
3.0             hashable    hash=3
'hi'            hashable    hash=<salted: differs every run>
(1, 2)          hashable    hash=-3550055125485641917
frozenset({1})  hashable    hash=-558064481276695278
None            hashable    hash=4238894112
True            hashable    hash=1
[1]             UNHASHABLE  (unhashable type: 'list')
{1: 2}          UNHASHABLE  (unhashable type: 'dict')
{1}             UNHASHABLE  (unhashable type: 'set')
(1, [2])        UNHASHABLE  (unhashable type: 'list')
```

Hashable	**Not hashable**
`int`, `float`, `complex`, `bool`, `str`, `bytes`, `None`, `frozenset`, `range`	`list`, `dict`, `set`, `bytearray`
`tuple` / `namedtuple` **if every item is hashable**	A tuple containing a list
Instances of your classes (by default hashed by identity)	Classes that define `__eq__` (the method behind `==`) without `__hash__` (the method behind `hash()`; Python sets `__hash__ = None`)
Frozen dataclasses (`@dataclass(frozen=True)`)	Regular dataclasses with `eq=True` (the default)

Equal values are the same key. `1 == 1.0 == True` and they hash equal, so:

```python
d = {1: "int one"}
d[1.0] = "float one"
d[True] = "bool true"
print(d)          # {1: 'bool true'}  — one key; the first key object is kept, value replaced
```

String hashes are randomised. For security (to stop attackers crafting many colliding keys — a "hash-flooding" denial of service), `str` and `bytes` hashes are salted with a random per-process seed (default since Python 3.3; controlled by the `PYTHONHASHSEED` environment variable — a setting handed to a program by the operating system). That's why the order of a set of strings can change between runs. Small ints hash to themselves (`hash(12345) == 12345`; `hash(-1)` is `-2` because `-1` is reserved as an error signal in C).

Mutating a key breaks the table. If a key's hash depends on mutable state and you change it while it's in a dict, the entry sits in the slot for the *old* hash:

```
lookup with same object: LOST
lookup with old value  : LOST
```

The same object can't be found (wrong slot), and a fresh object with the old value lands in the right slot but isn't `==`. That's precisely why built-in mutable types refuse to be hashable.

Big-O table (average case; worst case is O(n) when many keys collide)

Operation	dict	set
Lookup `d[k]` / `k in d` / `x in s`	O(1)	O(1)
Insert / update `d[k] = v`, `s.add(x)`	O(1) amortised	O(1) amortised
Delete `del d[k]`, `pop`, `s.remove(x)`, `discard`	O(1)	O(1)
`len()`	O(1)	O(1)
Iterate / `copy()` / `keys()`→list	O(n)	O(n)
`get`, `setdefault`, `popitem`	O(1)	`pop()` O(1)
`d1 | d2`, `update(other)`	O(len(other)) + copy	—
Union `s | t`	—	O(len(s) + len(t))
Intersection `s & t`	—	O(min(len(s), len(t)))
Difference `s - t`	—	O(len(s))
Symmetric difference `s ^ t`	—	O(len(s) + len(t))
`s <= t` (subset)	—	O(len(s))
Search a **value** `v in d.values()`	O(n)	—

The payoff, measured (`code/08_lookup_benchmark.py`, looking for the last element):

```
         n    list (µs)   set (µs)  dict (µs)
     1,000         5.26      0.030      0.031
    10,000        52.49      0.030      0.031
   100,000       536.93      0.030      0.031
 1,000,000      5533.07      0.030      0.032
```

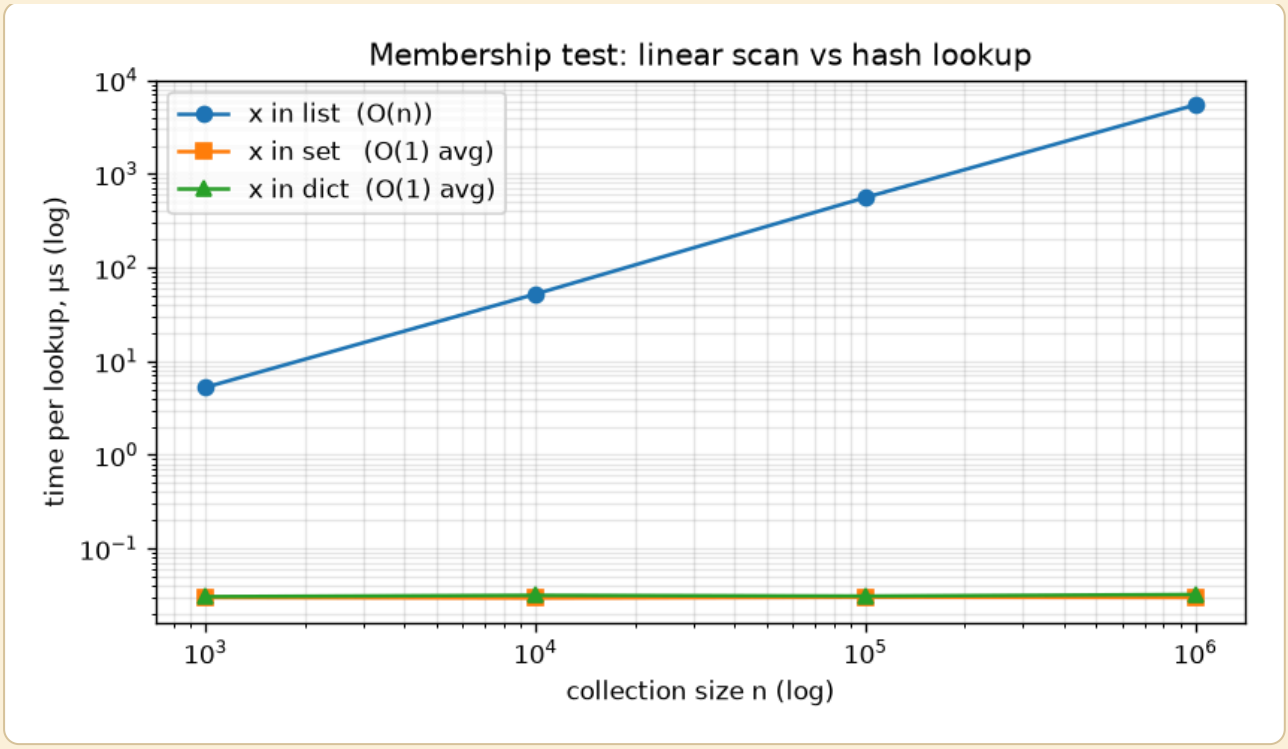


Figure — on a log–log plot the list line rises with slope 1 (linear); set and dict stay flat at ~30 ns (nanoseconds, billionths of a second). At a million items the set is ~180,000× faster.

Trade-offs

- **Memory:** a hash table keeps spare slots (up to 1/3 empty) plus stored hashes — much more memory than a list of the same items.
- **No positional access** and (for sets) no order.
- **Worst case** O(n) if many keys collide — rare with good hashes, but possible with a bad custom `__hash__` (e.g. `return 1`).

Common mistakes

- Defining `__eq__` on a class and being surprised it's no longer hashable — define `__hash__` from the same immutable fields (or use `@dataclass(frozen=True)`).
- Writing `__hash__` that uses mutable fields.
- Using floats as keys after arithmetic: `0.1 + 0.2` is not `0.3`, so it's a different key.

In the interview

30-second answer: "Dicts and sets are hash tables using open addressing. `hash(key)` picks a slot; collisions probe other slots; equality confirms the match. The table stays ≤ 2/3 full and resizes, so operations are O(1) average, O(n) worst. Keys must be hashable — their hash can't change — and equal objects must have equal hashes, which is why `1`, `1.0` and `True` are the same key."

- *Q: Why can't lists be dict keys?* A: They're mutable; if they changed, their hash would change and the entry would be lost.
- *Q: Why does set iteration order change between runs?* A: String hashes are randomly salted per process (hash-flooding protection).
- *Q: How are dicts ordered if they're hash tables?* A: Compact dict: the hash index points into an entries array kept in insertion order.

Self-check

1. Is `(1, [2])` hashable? → No — it contains a list.
2. What's the rule linking `==` and `hash`? → Equal objects must have equal hashes.
3. `len({1, 1.0, True})`? → 1.
4. Average cost of `x in big_set`? → O(1).

TL;DR

hash → slot → compare; ≤ 2/3 full; O(1) average. Hashable = hash never changes. Equal ⇒ same hash. String hashes are salted per run.

6.3 Nested dictionaries

What it is

A **nested dictionary** is a dict whose values are themselves dicts (or lists of dicts). It models **hierarchical** (levels within levels) data: a family tree, a config file, a JSON API response.

How it works

1. **Read** by chaining keys: `family["children"]["Alice"]["age"]`.
2. **Write** the same way: `family["children"]["Alice"]["age"] = 11`.
3. **Add a branch** by assigning a new dict at any level.
4. **Loop** over one level with `.items()`.

```
family = {
    "name": "John Doe",
    "children": {
        "Alice": {"age": 10, "gender": "Female"},
        "Bob": {"age": 8, "gender": "Male"},
    },
}
print("Parent:", family["name"])
print("Alice's age:", family["children"]["Alice"]["age"])
family["children"]["Alice"]["age"] = 11                  # update deep value
family["children"]["Cara"] = {"age": 3, "gender": "Female"}  # add a new branch
for child, info in family["children"].items():
    print(f"  {child}: {info['age']} ({info['gender']})")
```

Output:

```
Parent: John Doe
Alice's age: 10
  Alice: 11 (Female)
  Bob: 8 (Male)
  Cara: 3 (Female)
```

Safe access when a level may be missing:

```python
print(family.get("children", {}).get("Zed", {}).get("age", "unknown"))   # unknown

def deep_get(d, *keys, default=None):
    for k in keys:
        if not isinstance(d, dict) or k not in d:
            return default
        d = d[k]
    return d
print(deep_get(family, "children", "Bob", "age"), deep_get(family, "x", "y", default=-1))  # 8 -1
```

JSON round trip (convert to text and back) — nested dicts are exactly what `json` reads and writes:

```python
text = json.dumps(family, indent=2)
back = json.loads(text)
print(back == family)                                          # True
print(json.loads('{"1": [1, 2]}'), json.dumps({1: (1, 2)}))       # {'1': [1, 2]} {"1": [1, 2]}
```

JSON object keys are always strings and JSON has no tuples — so int keys come back as strings and tuples come back as lists.

Trade-offs

Nested dicts	Dataclasses / Pydantic models (classes from the Pydantic library that validate data)
Zero setup, matches JSON exactly	Named, typed fields, validation, IDE autocomplete
Typos in keys fail only at run time	Typos caught by type checker
Good for ad-hoc/dynamic data	Good for stable, well-known schemas (agreed shapes of the data)

When not to use it

For deeply nested, well-defined data used throughout a codebase, convert to dataclasses or Pydantic models at the boundary (where data enters or leaves your program); pass typed objects inside the program.

Common mistakes

- `KeyError` halfway down a chain — use `get(..., {})` chains, a helper like `deep_get`, or `try/except KeyError`.
- Shallow-copying a nested config and then mutating the copy's inner dict — changes the original too.
- Building nested structures by hand with many `if key not in d: d[key] = {}` checks — use `setdefault` or `defaultdict` (6.9).

Real-world examples

API responses (GitHub, Stripe) and config formats (JSON, YAML, TOML — human-friendly settings-file formats — via `tomllib`) all load as nested dicts; pandas' `json_normalize` flattens them into tables.

Self-check

1. How do you read Bob's gender? → `family["children"]["Bob"]["gender"]`.
2. What happens to tuple values in `json.dumps`? → They become JSON arrays (lists on load).
3. Safest one-liner for a possibly missing nested key? → Chained `.get(key, {})`.

> **TL;DR**
>
> Chain keys to read and write; guard missing levels with `get(..., {})`; nested dicts ≈ JSON.

6.4 Dictionary methods and view objects

The full method table

Method	What it does	Cost
`d.get(k, default=None)`	Value for `k`, or `default` — never raises	O(1)
`d.setdefault(k, default=None)`	Return `d[k]`; if missing, first insert `k: default`	O(1)
`d.update(other, **kw)`	Merge pairs from a dict / iterable of pairs / keywords, overwriting	O(len(other))
`d.pop(k[, default])`	Remove `k`, return its value; `KeyError` if absent and no default	O(1)
`d.popitem()`	Remove and return the **last inserted** `(k, v)` (LIFO — last in, first out); `KeyError` if empty	O(1)
`d.clear()`	Remove everything	O(n)
`d.copy()`	Shallow copy	O(n)
`dict.fromkeys(keys, value=None)`	**Class method**: new dict with every key mapped to the **same** `value`	O(n)
`d.keys()` / `d.values()` / `d.items()`	Live **view** objects	O(1) to create

Plus `len(d)`, `k in d`, `del d[k]`, `reversed(d)` (3.8+), `d | other`, `d |= other` (3.9+).

```python
info = {"name": "John", "age": 28, "city": "New York"}
print(info.get("name"), info.get("phone"))
info.update({"age": 29, "phone": "123-456-7890"})       # overwrite + add
info.update(zip_code="10001")                            # keyword form
print(info)
info.clear()
print(info)

new = dict.fromkeys(["a", "b", "c"], 1)
print(new)
print(new.pop("b"), new, new.items())
shared = dict.fromkeys(["x", "y"], [])                   # ONE list shared by all keys!
shared["x"].append(1)
print("fromkeys trap:", shared)
d = {"first": 1, "second": 2, "third": 3}
print(d.popitem(), d)                                    # LIFO: last inserted
```

Output:

```
John None
{'name': 'John', 'age': 29, 'city': 'New York', 'phone': '123-456-7890', 'zip_code': '10001'}
{}
{'a': 1, 'b': 1, 'c': 1}
1 {'a': 1, 'c': 1} dict_items([('a', 1), ('c', 1)])
fromkeys trap: {'x': [1], 'y': [1]}
('third', 3) {'first': 1, 'second': 2}
```

The `fromkeys` trap is the dict cousin of `[[0]*3]*3`: every key points at **one** list. Use `{k: [] for k in keys}` instead.

`get` vs `setdefault` vs `update` vs `|` — which one?

Want to…	Use	Changes `d`?
Read with a fallback	`d.get(k, default)`	No
Read, inserting a default if missing (e.g. grouping)	`d.setdefault(k, [])`	Only if missing
Merge another mapping into `d`	`d.update(other)` or `d |= other`	Yes
Get a **new** merged dict	`d | other` or `{**d, **other}`	No
Count / group many items	`Counter` / `defaultdict` (6.9)	—

```python
words = ["apple", "avocado", "banana", "blueberry", "cherry"]
by_letter = {}
for w in words:
    by_letter.setdefault(w[0], []).append(w)            # insert [] only if missing
print(by_letter)
counts = {}
for ch in "hello":
    counts[ch] = counts.get(ch, 0) + 1                  # get with default for counting
print(counts)
```

Output:

```
{'a': ['apple', 'avocado'], 'b': ['banana', 'blueberry'], 'c': ['cherry']}
{'h': 1, 'e': 1, 'l': 2, 'o': 1}
```

Note that `setdefault`'s default is **evaluated every call** even when the key exists (`[]` is built and thrown away) — harmless for `[]`, costly for expensive defaults. `defaultdict` avoids that.

View objects

`keys()`, `values()` and `items()` return **views**: lightweight windows onto the dict that reflect later changes. Key views (and item views, if values are hashable) also support set operations.

```python
stock = {"apple": 3, "pear": 0}
k, v, it = stock.keys(), stock.values(), stock.items()
stock["fig"] = 7
print(k, v, it, sep="\n")
print(k & {"apple", "kiwi"}, sorted(k - {"pear"}))      # keys views support set ops
print(sorted(stock, key=stock.get, reverse=True))       # keys sorted by value
print(max(stock.items(), key=lambda kv: kv[1]))
```

Output:

```
dict_keys(['apple', 'pear', 'fig'])
dict_values([3, 0, 7])
dict_items([('apple', 3), ('pear', 0), ('fig', 7)])
{'apple'} ['apple', 'fig']
['fig', 'apple', 'pear']
('fig', 7)
```

Common mistakes

- `dict.fromkeys(keys, [])` sharing one mutable value.
- `d.update(k=v)` when `k` is a variable — the keyword form uses the literal name `"k"`. Use `d[k] = v`.
- Assuming `popitem()` removes a random item — since 3.7 it's always the last inserted.

In the interview

30-second answer: "`get` reads with a default, `setdefault` reads-or-inserts, `update` / `|=` merge in place, `|` returns a new merged dict with the right side winning. `pop` removes by key, `popitem` removes the last inserted. `keys/values/items` are live views, and key views support set operations."

Self-check

1. `{"a": 1}.setdefault("b", 5)` — return value and resulting dict? → `5`; `{'a': 1, 'b': 5}`.
2. What does `popitem()` remove? → The most recently inserted pair.
3. Keys common to two dicts? → `d1.keys() & d2.keys()`.

TL;DR

`get` (read), `setdefault` (read-or-insert), `update` / `|=` (merge in place), `|` (merge new), `pop` / `popitem`, views are live and set-like. Avoid `fromkeys` with mutable values.

6.5 Dict and set comprehensions

What it is

The comprehension syntax from Chapter 5 also builds dicts and sets:

```
{key_expr: value_expr for item in iterable if condition}   # dict
{expr for item in iterable if condition}                   # set
```

How it works

```
words = ["apple", "banana", "cherry"]
print({w: len(w) for w in words})
prices = {"apple": 1.2, "banana": 0.5, "cherry": 3.0}
print({k: v for k, v in prices.items() if v >= 1})
print({v: k for k, v in {"a": 1, "b": 2}.items()})    # invert
print(sorted({c for c in "mississippi"}))              # set comprehension
```

Output:

```
{'apple': 5, 'banana': 6, 'cherry': 6}
{'apple': 1.2, 'cherry': 3.0}
{1: 'a', 2: 'b'}
['i', 'm', 'p', 's']
```

- Filtering a dict: iterate `.items()` and keep pairs that pass the test.
- **Inverting** a dict swaps keys and values — if two keys share a value, the later one wins (values must also be hashable).
- `{}` is an **empty dict**, not an empty set. Write `set()`.

Common mistakes

- Duplicate keys in a dict comprehension silently overwrite earlier ones.
- Using a comprehension when `dict(zip(keys, values))` is simpler.

Self-check

1. `{x % 3 for x in range(10)}`? → `{0, 1, 2}`.
2. Build `{name: len(name)}` for a list `names`? → `{n: len(n) for n in names}`.

TL;DR

`{k: v for ...}` for dicts, `{x for ...}` for sets; `{}` alone is a dict.

6.6 Sets — unique, unordered collections

What it is

A **set** is an unordered, mutable collection of **unique, hashable** items: `{"apple", "banana"}` or `set(iterable)`. Duplicates are dropped automatically. There's no indexing (`s[0]` is a `TypeError`) because there's no position.

Analogy: a bag of distinct marbles. You can ask "is the red one in the bag?" instantly, add or take out marbles, but they aren't in any order.

Why it matters

- **De-duplication** in one step: `set(emails)`.
- **Fast membership** (O(1) average, 6.2): "have I seen this before?" in crawlers (programs that visit web pages by following links), graph searches, validation.
- **Set algebra** answers questions like "which users are in both groups?" in one operator (6.7).

How it works — define, add, update, remove, copy, loop

```
fruits = {"apple", "banana", "cherry", "apple"}
colors = set(["red", "blue", "red", "green"])
print(sorted(fruits), sorted(colors), len(fruits))   # sorted() only for stable printing
empty_set, empty_dict = set(), {}
print(type(empty_set).__name__, type(empty_dict).__name__)
print(set("hello"), set(range(3)))                    # str order varies run to run

nums = {1, 2, 3}
nums.add(4); nums.add(2)                  # adding an existing item: no effect
nums.update([5, 6], (7,))                 # any iterables
print(nums)
nums.remove(1)
nums.discard(60)                          # absent: silently ignored
try:
    nums.remove(60)
except KeyError as e:
    print("KeyError:", e)
backup = nums.copy()
nums.clear()
print(nums, backup)
print("pop() ->", backup.pop(), backup)    # removes an ARBITRARY element
```

Output:

```
['apple', 'banana', 'cherry'] ['blue', 'green', 'red'] 3
set dict
{'l', 'h', 'e', 'o'} {0, 1, 2}          <- letter order varies from run to run
{1, 2, 3, 4, 5, 6, 7}
KeyError: 60
set() {2, 3, 4, 5, 6, 7}
pop() -> 2 {3, 4, 5, 6, 7}
```

- `add(x)` inserts one item; `update(*iterables)` inserts many.
- `remove(x)` raises `KeyError` if `x` is missing; `discard(x)` doesn't — use it when absence is fine.
- `pop()` removes and returns an **arbitrary** element (whatever the table yields first) — don't rely on which.
- `copy()` gives an independent set (its items are immutable anyway, so shallow = full here).
- Small integers often *appear* sorted because they hash to themselves; that's a coincidence of the implementation, not a guarantee.

De-duplicating while keeping order — a set loses order, so use a dict:

```
items = ["b", "a", "b", "c", "a"]
print(list(dict.fromkeys(items)))       # ['b', 'a', 'c']
```

Common mistakes

- `{}` for an empty set (it's a dict).
- `set("hello")` when you meant `{"hello"}` — the first splits into characters.
- Putting lists or dicts in a set → `TypeError: unhashable type`.
- Relying on set order for output — sort it.

Self-check

1. `len({1, 2, 2, 3})`? → 3.

2. `remove` vs `discard`? → `remove` raises `KeyError` if absent; `discard` doesn't.
3. How to dedupe a list but keep first-seen order? → `list(dict.fromkeys(lst))`.

TL;DR

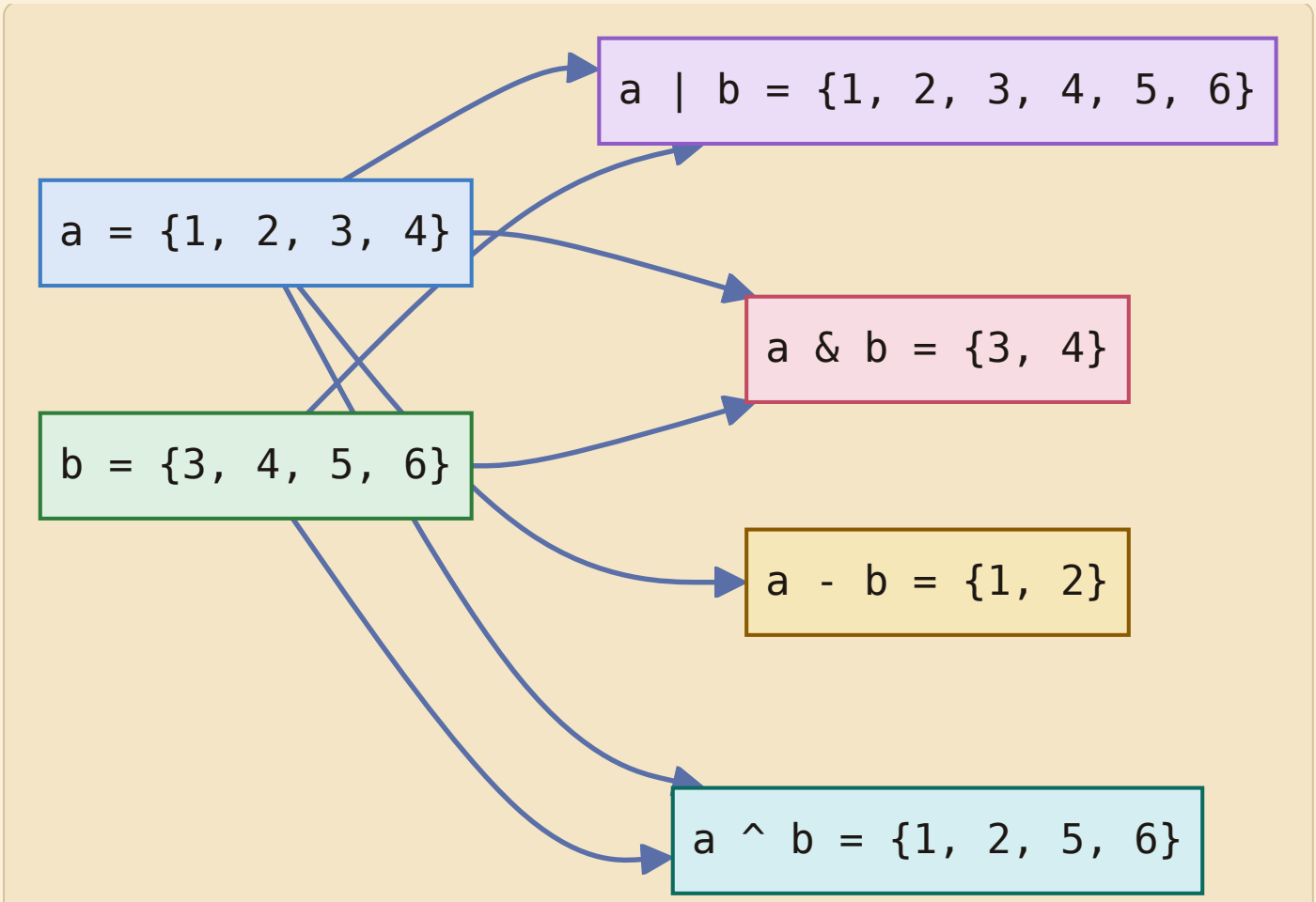
Unique, unordered, hashable items; `add`/`update`, `remove` (raises) / `discard` (silent); `set()` for empty; O(1) membership.

6.7 Set operations

What it is

Mathematical set algebra, available both as **operators** (both sides must be sets/frozensets) and as **methods** (accept any iterable).

Operation	Operator	Method	Result
Union — in either	`a | b`	`a.union(b, ...)`	Everything from both
Intersection — in both	`a & b`	`a.intersection(b, ...)`	Common items
Difference — in `a` not `b`	`a - b`	`a.difference(b, ...)`	`a` minus `b`
Symmetric difference — in exactly one	`a ^ b`	`a.symmetric_difference(b)`	Union minus intersection
Subset	`a <= b`	`a.issubset(b)`	Every item of `a` is in `b`
Proper subset	`a < b`	—	Subset and not equal
Superset	`a >= b`	`a.issuperset(b)`	Every item of `b` is in `a`
Disjoint	—	`a.isdisjoint(b)`	No common items




```
a, b = {1, 2, 3, 4}, {3, 4, 5, 6}
print("union        a | b :", a | b, a.union(b))
print("intersection a & b :", a & b, a.intersection(b))
print("difference   a - b :", a - b, a.difference(b), "  b - a:", b - a)
print("sym. diff    a ^ b :", a ^ b, a.symmetric_difference(b))
print("method accepts any iterable:", a.union([9, 10]), " operator needs sets")
try:
    a | [9]
except TypeError as e:
    print("TypeError:", e)
print("multi-arg:", a.intersection(b, {4, 99}))

small, big, other = {1, 2}, {1, 2, 3}, {7}
print(small <= big, small.issubset(big), small < big, big >= small, big.issuperset(small))
print(small.isdisjoint(other), small < small, small <= small)
```

Output:

```
union        a | b : {1, 2, 3, 4, 5, 6} {1, 2, 3, 4, 5, 6}
intersection a & b : {3, 4} {3, 4}
difference   a - b : {1, 2} {1, 2}   b - a: {5, 6}
sym. diff    a ^ b : {1, 2, 5, 6} {1, 2, 5, 6}
method accepts any iterable: {1, 2, 3, 4, 9, 10}  operator needs sets
TypeError: unsupported operand type(s) for |: 'set' and 'list'
multi-arg: {4}
True True True True True
True False True
```

Difference is **not symmetric**: `a - b = {1, 2}` but `b - a = {5, 6}`. A set is a subset of itself (`<=`) but not a proper subset (`<`).

Trade-offs

Operators are shorter and make you convert inputs explicitly (catching type mistakes); methods are more flexible (any iterable, several arguments at once).

Real-world examples

- Permissions: `required <= user_permissions`.
- Diffing: `new_ids - old_ids` (added), `old_ids - new_ids` (removed) when syncing records.
- Recommendations: "people who liked X also liked" via intersections; Jaccard similarity (shared items ÷ all distinct items) = `len(a & b) / len(a | b)`, used in near-duplicate detection.

In the interview

> **30-second answer:** "`|` union, `&` intersection, `-` difference, `^` symmetric difference, `<=` / `>=` subset/superset, `isdisjoint`. Operators need sets; methods accept any iterable. Intersection costs O(min(len a, len b)) because it iterates the smaller set and probes the larger."

Self-check

1. `{1, 2, 3} ^ {2, 3, 4}`? → `{1, 4}`.
2. `{1, 2} < {1, 2}`? → `False`.
3. Users in group A but not B? → `A - B`.

TL;DR

`| & - ^` plus subset/superset/disjoint; methods take any iterable; difference is one-directional.

6.8 Set methods (complete table) and `frozenset`

All set methods

Method	What it does	Operator form	Returns
`add(x)`	Add one element (no effect if present)	—	`None`
`update(*its)`	Add all elements from iterables	`|=`	`None`
`remove(x)`	Remove `x`; `KeyError` if absent	—	`None`
`discard(x)`	Remove `x` if present	—	`None`
`pop()`	Remove and return an arbitrary element; `KeyError` if empty	—	element
`clear()`	Remove everything	—	`None`
`copy()`	Shallow copy	—	new set
`union(*its)`	All elements	`|`	new set
`intersection(*its)`	Common elements	`&`	new set
`intersection_update(*its)`	Keep only common elements	`&=`	`None`
`difference(*its)`	Elements not in the others	`-`	new set
`difference_update(*its)`	Remove elements found in the others	`-=`	`None`
`symmetric_difference(it)`	In exactly one	`^`	new set
`symmetric_difference_update(it)`	Update to "in exactly one"	`^=`	`None`
`issubset(it)`	All my elements are in `it`	`<=`	bool
`issuperset(it)`	All of `it`'s elements are in me	`>=`	bool
`isdisjoint(it)`	No common elements	—	bool

The `_update` methods change the set **in place** and return `None`; the plain ones return a **new** set.

```
s = {1, 2, 3, 4}
s.intersection_update({2, 3, 9});         print("&= :", s)
s.difference_update({3});                 print("-= :", s)
s.symmetric_difference_update({2, 8});    print("^= :", s)
s |= {100};                               print("|= :", s)
```

Output:

```
&= : {2, 3}
-= : {2}
^= : {8}
|= : {8, 100}
```

Each `_update` method has an augmented-operator (operation plus assignment, like `+=`) twin (`&=`, `-=`, `^=`, `|=`); the operator forms require sets on both sides, the methods accept any iterables.

`frozenset` — the immutable set

What it is

A **frozenset** is a set that can't be changed after creation. It has all the non-mutating methods (`union`, `issubset`, …) and none of the mutating ones (`add`, `remove`, `update`, …).

Why it matters

Because it's immutable, it's **hashable**: it can be a dict key or an element of another set — something a normal set can never be.

```
fs = frozenset([1, 2, 3])
try:
    fs.add(4)
except AttributeError as e:
    print("AttributeError:", e)
groups = {frozenset({"alice", "bob"}): "team A"}   # usable as a dict key
print(groups[frozenset({"bob", "alice"})])
print(fs | {9}, type(fs | {9}).__name__)            # result type follows the left operand
```

Output:

```
AttributeError: 'frozenset' object has no attribute 'add'
team A
frozenset({1, 2, 3, 9}) frozenset
```

The lookup works with the members in a different order because sets compare (and hash) by contents, not order — handy for keys like "unordered pair of users".

Common mistakes

- Expecting `s.union(t)` to modify `s` (it returns a new set) — or expecting `s.update(t)` to return the result (it returns `None`).
- Using `pop()` as if it returned the smallest element.

Self-check

1. Which returns `None`: `intersection` or `intersection_update`? → `intersection_update`.
2. Can a set contain another set? → Only a `frozenset`.
3. Key for an undirected edge (a connection between two points with no direction) `{u, v}`? → `frozenset((u, v))`.

TL;DR

Plain methods return new sets, `_update` methods mutate; `frozenset` = immutable, hashable set for keys and nested sets.

6.9 The `collections` module

The standard library's `collections` module supplies specialised containers that fix the most common "plain dict/list isn't quite right" situations.

`Counter` — counting hashable things

What it is: a dict subclass (a specialised version of `dict`) mapping item → count. Missing items count as `0` (no `KeyError`).

```
words = "the cat and the hat and the bat".split()
c = Counter(words)
print(c)
print(c.most_common(2), c["the"], c["dog"])          # missing key -> 0, no KeyError
c.update(["cat", "cat"]); c.subtract(["the"])
print(c.most_common(3))
a, b = Counter("aabbb"), Counter("abccc")
print(a + b, a - b, a & b, a | b, sep=" | ")
print(Counter("mississippi").total())                # 3.10+
```

Output:

```
Counter({'the': 3, 'and': 2, 'cat': 1, 'hat': 1, 'bat': 1})
[('the', 3), ('and', 2)] 3 0
[('cat', 3), ('the', 2), ('and', 2)]
Counter({'b': 4, 'a': 3, 'c': 3}) | Counter({'b': 2, 'a': 1}) | Counter({'a': 1, 'b': 1}) |
Counter({'b': 3, 'c': 3, 'a': 2})
11
```

- `most_common(n)` — top-n pairs (ties keep first-seen order).
- `update` **adds** counts (unlike `dict.update`, which replaces); `subtract` subtracts and may leave zero or negative counts.
- `+` / `-` add/subtract and **drop** results ≤ 0; `&` takes the minimum per key, `|` the maximum.
- `total()` sums all counts (3.10+).

Real-world: word frequencies, log-level tallies, top-k error messages, the "bag of words" (text represented as word counts) step in text classification (sorting documents into categories).

`defaultdict` — dict with automatic defaults

What it is: a dict subclass that calls a **factory** (a function that builds new values: `list`, `int`, `set`, any zero-argument callable — anything you can call with `()`) to create a value the first time a missing key is **accessed with** `[]`.

```
groups = defaultdict(list)
for kind, name in [("fruit", "apple"), ("veg", "kale"), ("fruit", "fig")]:
    groups[kind].append(name)                        # missing key -> list() created
print(dict(groups))
counts = defaultdict(int)
for ch in "banana":
    counts[ch] += 1
print(dict(counts))
print("veg" in groups, "meat" in groups, groups.get("meat"))   # get/in don't create keys
_ = groups["meat"]                                               # [] access DOES
print("after groups['meat']:", dict(groups))
tree = lambda: defaultdict(tree)                                 # autovivifying tree
t = tree(); t["a"]["b"]["c"] = 1
```

Output:

```
{'fruit': ['apple', 'fig'], 'veg': ['kale']}
{'b': 1, 'a': 3, 'n': 2}
True False None
after groups['meat']: {'fruit': ['apple', 'fig'], 'veg': ['kale'], 'meat': []}
```

Watch out: **just reading** `d[missing]` inserts an empty value — a common source of mysterious extra keys. Use `in` or `.get()` for read-only checks.

Real-world: adjacency lists (for each node, the list of its neighbours) for graphs (`defaultdict(list)`), inverted indexes (word → the documents containing it, as in search engines; `defaultdict(set)` word → document ids), grouping rows by a column.

`OrderedDict` — a dict with reordering tools

What it is: the pre-3.7 way to get an ordered dict. Plain dicts are now ordered too, so `OrderedDict` matters only for its extras:

- `move_to_end(key, last=True)` — O(1) move to the end (or front with `last=False`).
- `popitem(last=False)` — pop from the **front** (FIFO — first in, first out); plain `dict.popitem()` only pops the end.
- **Order-sensitive equality**: two `OrderedDict`s are equal only if the order matches too.

```
od = OrderedDict.fromkeys("abcde")
od.move_to_end("b")                  # to the end
od.move_to_end("e", last=False)      # to the front
print(list(od))                      # ['e', 'a', 'c', 'd', 'b']
print(od.popitem(last=False))        # ('e', None)
print(OrderedDict(a=1, b=2) == OrderedDict(b=2, a=1), dict(a=1, b=2) == dict(b=2, a=1))  # False True
```

The classic use — an LRU (least-recently-used) cache:

```
class LRUCache:
    def __init__(self, capacity):
        self.capacity, self.data = capacity, OrderedDict()
    def get(self, key):
        if key not in self.data:
            return None
        self.data.move_to_end(key)           # mark as most recently used
        return self.data[key]
    def put(self, key, value):
        self.data[key] = value
        self.data.move_to_end(key)
        if len(self.data) > self.capacity:
            self.data.popitem(last=False)    # evict least recently used

cache = LRUCache(2)
cache.put("a", 1); cache.put("b", 2); cache.get("a"); cache.put("c", 3)
print("LRU keeps:", list(cache.data))        # LRU keeps: ['a', 'c']
```

Reading `"a"` made it recent, so adding `"c"` evicted (removed to make room for) `"b"`. For caching function results, prefer the built-in `@functools.lru_cache`.

`deque` — double-ended queue

What it is: a sequence with **O(1)** `append`/`pop` at **both** ends (a list is O(n) at the front). Implemented as a doubly-linked list (a chain where each block links to the next and the previous) of fixed-size blocks, so indexing in the middle is O(n).

```
q = deque([1, 2, 3])
q.append(4); q.appendleft(0)
print("after append/appendleft:", q)
left, right = q.popleft(), q.pop()
print("popleft ->", left, "| pop ->", right, "| now", q)
q.rotate(1); print("rotate(1):", q)
last3 = deque(maxlen=3)
for x in range(6):
    last3.append(x)                  # old items fall off the other end
print("maxlen=3:", last3)
```

Output:

```
after append/appendleft: deque([0, 1, 2, 3, 4])
popleft -> 0 | pop -> 4 | now deque([1, 2, 3])
rotate(1): deque([3, 1, 2])
maxlen=3: deque([3, 4, 5], maxlen=3)
```

- `maxlen` gives a bounded buffer (a holding area with a fixed maximum size) — perfect for "last N lines/events" (like `tail -n`, the Unix command that shows a file's last lines).
- `append` and `popleft` are thread-safe (safe to call from several threads at once) in CPython, which is why deques back simple producer/consumer queues (one part of a program adds work, another takes it) (`queue.Queue` adds blocking — waiting until an item arrives — and locking on top).

Real-world: BFS (breadth-first search: exploring a graph level by level) queues in graph search, sliding-window algorithms (working over a moving range of the most recent items), undo history with a cap, rate limiters (which cap how many requests are allowed per time period) that keep recent timestamps.

`ChainMap` — layered lookups

What it is: a view over several mappings searched **in order**; the first one containing the key wins. Nothing is copied.

```
defaults = {"color": "red", "user": "guest", "debug": False}
env = {"user": "admin"}
cli = {"debug": True}
cfg = ChainMap(cli, env, defaults)    # first mapping wins
print(cfg["color"], cfg["user"], cfg["debug"])
cfg["color"] = "blue"                 # writes go to the FIRST mapping only
print(cli, defaults["color"])
child = cfg.new_child({"user": "tmp"})
print(child["user"], cfg["user"])
```

Output:

```
red admin True
{'debug': True, 'color': 'blue'} red
tmp admin
```

Real-world: configuration precedence (which source wins: command-line > environment variables > defaults), nested scopes (regions where names are visible) in interpreters/template engines (`new_child()` pushes a scope). The standard library's `string.Template.substitute` uses a `ChainMap` internally to combine a mapping with keyword arguments.

`namedtuple` — tuple with field names

Covered fully in Chapter 5 (5.13): `Card = namedtuple("Card", "rank suit")` → `Card("A", "spades").rank == "A"`, with `_fields`, `_asdict()`, `_replace()`.

```
A Card(rank='10', suit='hearts') ('rank', 'suit')
```

Common mistakes (collections)

- Using `Counter.update` expecting replacement semantics (behaviour).
- Accidentally creating keys in a `defaultdict` by reading.
- Using `list.pop(0)` instead of `deque.popleft()` for queues.
- Using `OrderedDict` just for ordering on Python 3.7+ — a plain dict already does it.

In the interview

30-second answer: "`Counter` counts and gives `most_common`; `defaultdict` auto-creates values for grouping; `deque` has O(1) appends and pops at both ends — use it for queues and bounded buffers; `OrderedDict` adds `move_to_end` and FIFO `popitem`, which makes an LRU cache easy; `ChainMap` layers mappings for config precedence; `namedtuple` gives tuples field names."

- *Q: Implement an LRU cache?* A: `OrderedDict` with `move_to_end` on access and `popitem(last=False)` on overflow — both O(1).
- *Q: Why deque for BFS?* A: `popleft()` is O(1); `list.pop(0)` is O(n).
- *Q: `Counter` + vs `update`?* A: `+` returns a new Counter and drops non-positive counts; `update` adds in place and keeps them.

Self-check

1. `Counter("aab")["z"]`? → `0`.
2. Build a graph's adjacency lists without key checks? → `defaultdict(list)`.
3. Keep only the last 100 log lines? → `deque(maxlen=100)`.
4. Which mapping receives writes in a `ChainMap`? → The first one.

TL;DR

Counter = counts; defaultdict = auto-values; OrderedDict = reordering + FIFO pop (LRU); deque = fast both ends + maxlen; ChainMap = layered lookup; namedtuple = named tuple fields.

6.10 `heapq` — priority queues on a plain list

What it is

A **binary min-heap** is a list arranged so that `heap[k] <= heap[2k+1]` and `heap[k] <= heap[2k+2]` for every `k`. The smallest item is always at `heap[0]`. The `heapq` module provides functions that keep a normal list in heap order.

Why it matters

Whenever you repeatedly need "the smallest (or most urgent) item next" — task schedulers, Dijkstra's shortest path (the classic algorithm for the cheapest route through a weighted network), merging sorted streams, top-k (the k best items) — a heap does it in O(log n) per operation instead of re-sorting.

How it works

Function	Does	Cost
`heapify(list)`	Rearrange a list into a heap in place	O(n)
`heappush(h, x)`	Add `x` (sift up — move it up until heap order holds)	O(log n)
`heappop(h)`	Remove and return the smallest (sift down)	O(log n)
`h[0]`	Peek at the smallest	O(1)
`heappushpop(h, x)` / `heapreplace(h, x)`	Push then pop / pop then push, efficiently	O(log n)
`nsmallest(k, it)` / `nlargest(k, it)`	Top-k without full sort	O(n log k)
`merge(*sorted_its)`	Lazily merge sorted inputs	O(n log m)
`heapify_max`, `heappush_max`, `heappop_max`, …	Max-heap versions (**new in Python 3.14**)	as above

```
def heapify_neg(xs):                    # helper: store negatives -> behaves as a max-heap
    h = [-x for x in xs]
    heapq.heapify(h)
    return h

h = [5, 1, 8, 3, 2]
heapq.heapify(h)                        # O(n), in place; h[0] is the smallest
print("heapified:", h, "min:", h[0])
heapq.heappush(h, 0)
print("pops:", [heapq.heappop(h) for _ in range(3)], "left:", h)
print(heapq.nsmallest(2, [7, 3, 9, 1]), heapq.nlargest(2, [7, 3, 9, 1]))
print("max-heap trick (negate):", -heapq.heappop(heapify_neg([3, 9, 4])))
```

Output:

```
heapified: [1, 2, 8, 3, 5] min: 1
pops: [0, 1, 2] left: [3, 5, 8]
[1, 3] [9, 7]
max-heap trick (negate): 9
```

The heapified list `[1, 2, 8, 3, 5]` is **not sorted** — only the heap property holds (1 ≤ 2, 1 ≤ 8, 2 ≤ 3, 2 ≤ 5). Before 3.14, the standard trick for a max-heap is to push negated numbers.

Priority queue with tie-breaking:

```python
tasks, counter = [], 0
for prio, name in [(2, "write docs"), (1, "fix prod bug"), (3, "refactor"), (1, "reply to CEO")]:
    heapq.heappush(tasks, (prio, counter, name))   # (priority, sequence, payload)
    counter += 1
while tasks:
    prio, _, name = heapq.heappop(tasks)
    print(f"  p{prio}: {name}")
```

Output:

```
  p1: fix prod bug
  p1: reply to CEO
  p2: write docs
  p3: refactor
```

Tuples compare element by element, so priority decides first; the increasing counter breaks ties in insertion order **and** stops Python from ever comparing the payloads (the actual data carried — here the task names; they might not be comparable).

`heapq.merge([1, 4, 9], [2, 3, 10], [5])` → `[1, 2, 3, 4, 5, 9, 10]` — the core of external sorting (merging sorted files too big for memory).

Trade-offs

Need	Use
Smallest item repeatedly, with inserts	`heapq`
Just the top k once	`heapq.nlargest/nsmallest` (or `sorted(...)[:k]` if k is close to n)
Only min or max once	`min()` / `max()` — O(n)
Thread-safe blocking priority queue	`queue.PriorityQueue` (uses heapq inside)
Sorted order at all times with lookups	`bisect` on a list, or a sorted-container library

Real-world examples

`asyncio`'s event loop (the scheduler that runs asynchronous tasks) keeps scheduled callbacks (functions to be called later; `call_later`) in a heap; the `sched` module and `queue.PriorityQueue` are built on `heapq`; Dijkstra/A* (a faster shortest-path search guided by a distance estimate) implementations in Python almost always use it.

Self-check

1. Where's the smallest item in a heap? → `h[0]`.
2. Cost of `heapify` vs n pushes? → O(n) vs O(n log n).
3. Why add a counter to heap entries? → Stable tie-breaking and to avoid comparing payloads.

TL;DR

`heapify` O(n), push/pop O(log n), `h[0]` is the min; tuples `(priority, count, item)`; `nlargest` for top-k; max-heap = negate (or 3.14's `_max` functions).

6.11 `bisect` — binary search on sorted lists

What it is

`bisect` finds where a value belongs in an **already sorted** list using binary search — O(log n) comparisons.

How it works

- `bisect_left(a, x)` — first index where `x` could be inserted keeping order (before any equal items).
- `bisect_right(a, x)` / `bisect(a, x)` — index **after** any equal items.
- `insort(a, x)` (= `insort_right`) — insert keeping order: O(log n) search **+ O(n)** list shift.
- Since Python 3.10, all functions accept `key=`.

Worked example on `[10, 20, 20, 30, 40]` searching for 20: the middle element (index 2) is 20; `bisect_left` keeps searching left and lands at index 1; `bisect_right` lands after the last 20 at index 3.

```
scores = [10, 20, 20, 30, 40]
print(bisect.bisect_left(scores, 20), bisect.bisect_right(scores, 20), bisect.bisect(scores, 25))
bisect.insort(scores, 25)
print(scores)

def contains(sorted_list, x):
    i = bisect.bisect_left(sorted_list, x)
    return i < len(sorted_list) and sorted_list[i] == x
print(contains(scores, 30), contains(scores, 31))

def grade(score, cutoffs=(60, 70, 80, 90), letters="FDCBA"):
    return letters[bisect.bisect(cutoffs, score)]
print([grade(s) for s in (33, 60, 77, 89, 90, 100)])

people = [("Ann", 25), ("Bob", 31), ("Cid", 40)]           # sorted by age
i = bisect.bisect_left(people, 30, key=lambda p: p[1])      # key= is 3.10+
print("first person aged >= 30:", people[i])
```

Output:

```
1 3 3
[10, 20, 20, 25, 30, 40]
True False
['F', 'D', 'C', 'B', 'A', 'A']
first person aged >= 30: ('Bob', 31)
```

The grading function is the classic example from the Python docs: `bisect` turns a table of cutoffs into a lookup — a score of exactly 60 lands to the **right** of the 60 cutoff, i.e. "D".

Trade-offs

- Searching is O(log n), but `insort` is O(n) because the list must shift. For many inserts plus lookups on big data, a set/dict (if you don't need order) or a balanced tree (a self-adjusting sorted tree with O(log n) inserts)/`sortedcontainers.SortedList` (third-party) is better.
- The list **must** be sorted by the same key you search with — bisect doesn't check.

Real-world examples

`random.choices` uses `bisect` on cumulative weights (running totals of the weights) to pick weighted samples; time-series (data points ordered by time) code uses it to find the first event after a timestamp; percentile/bucket lookups (tax brackets, grade cutoffs) are one call.

Self-check

1. `bisect_left([1, 2, 2, 3], 2)`? → `1`. `bisect_right`? → `3`.
2. Cost of `insort` on a list of n items? → O(n) (O(log n) search + O(n) shift).
3. What must be true of the list? → Sorted (by the same key).

TL;DR

Binary search on sorted lists: `bisect_left/right` O(log n), `insort` O(n), `key=` since 3.10.

6.12 When to use what — the decision table

You need…	Use	Why	Key costs
Ordered items, index access, append at end	`list`	Dynamic array	index O(1), append O(1)*, `in` O(n)
Fixed record / dict key / return several values	`tuple` / `namedtuple`	Immutable, hashable, light	index O(1)
Look up values by key	`dict`	Hash table	get/set/del O(1) avg
Fast membership / uniqueness	`set`	Hash table	`in` O(1) avg
A set as a key or inside a set	`frozenset`	Immutable, hashable	same as set
Count things / top-k frequencies	`Counter`	Counting dict with `most_common`	update O(k)
Group into lists/sets without key checks	`defaultdict(list/set)`	Auto-created values	O(1)
Queue / stack at both ends / last N items	`deque` (`maxlen=N`)	O(1) at both ends	append/pop O(1), middle index O(n)
LRU cache / reorder keys / FIFO pop	`OrderedDict` (or `functools.lru_cache`)	`move_to_end`, `popitem(last=False)`	O(1)
Layered config / scopes	`ChainMap`	Search mappings in order, no copying	lookup O(number of maps)
Repeatedly take the smallest/most urgent	`heapq`	Binary heap	push/pop O(log n), peek O(1)
Top-k from a large iterable	`heapq.nlargest/nsmallest`	Avoids a full sort	O(n log k)
Search/insert in a sorted list	`bisect`	Binary search	search O(log n), insert O(n)
Dedupe but keep order	`dict.fromkeys(items)`	Dicts keep insertion order	O(n)
Compact numeric storage / fast maths	`array.array` / NumPy	Chapter 5	—

(* amortised)

In the interview — putting it together

Prompt: "Find the 3 most frequent words in a log file and the first line where each appears."

Answer sketch: stream lines; `Counter` for counts; a dict with `setdefault(word, line_no)` for first occurrence; `counter.most_common(3)` at the end. Everything is O(total words) plus O(u log 3) for the top-k — no nested loops, no `list.index`.

Chapter TL;DR

Topic	Remember
Dicts	Key → value, O(1) avg, insertion-ordered since 3.7; `[]` raises `KeyError`, `get` defaults; `|` merges (3.9+); `copy` is shallow; don't resize while iterating.
Hashing	hash → slot → compare; ≤ 2/3 full; equal ⇒ same hash; keys must be immutable/hashable; string hashes salted per run.
Nested dicts	Chain keys; guard with `get(..., {})`; map 1:1 to JSON (keys become strings).
Dict methods	`get`, `setdefault`, `update` / `|=`, `pop`, `popitem` (LIFO), `fromkeys` (shared value trap), live views with set ops.
Comprehensions	`{k: v for ...}`, `{x for ...}`; `{}` is a dict.
Sets	Unique, unordered, hashable items; `add` / `update`; `remove` raises, `discard` doesn't; `set()` for empty.
Set operations	`| & - ^`, `<= < >= >`, `isdisjoint`; methods accept iterables; `_update` versions mutate.
frozenset	Immutable, hashable set — keys and sets of sets.
collections	Counter, defaultdict, OrderedDict (LRU), deque (queues, maxlen), ChainMap (layers), namedtuple.
heapq	Min-heap on a list; push/pop O(log n); `(prio, count, item)`; `nlargest`; max-heap functions in 3.14.
bisect	Binary search on sorted lists; `insort` is O(n); `key=` since 3.10.